Roughness of biopores and cracks in Bt-horizons assessed by confocal laser scanning microscopy

Martin Leue[1]* and **Horst H. Gerke**[1]

[1] Institute of Soil Landscape Research, Leibniz-Centre for Agricultural Landscape Research (ZALF), 15374 Müncheberg, Germany

Abstract

In structured soils, water and reactive solutes can preferentially move through larger inter-aggregate pores, cracks, and biopores. The surface roughness of such macropores is crucial for describing microbial habitats and the exchange of water and solutes between macropores and the soil matrix together with other properties. The objective of this study was to compare the roughness of intact structural surfaces from the Bt-horizons of five Luvisols developed on loess and glacial till and to test the applicability of confocal laser scanning microscopy. Samples of 5 to 10 cm edge length with intact structural surfaces including cracks with and without clay-organic coatings, earthworm burrow walls, and root channels were prepared manually. The surface roughness of these structures was determined with a confocal laser scanning microscope of the type Keyence VK-X100K. The root-mean-squared roughness (R_q) the curvature (R_{cu}) and the ratio between surface area and base area (R_A) were calculated from selected surface regions of interest of 0.342 mm^2 with an elevation resolution of 0.02 µm. The roughness was smaller for coated as compared to uncoated cracks and earthworm burrows of the Bt-horizons. This reduction of roughness by the illuviation of clayey material was similar for the structural surfaces of the coarser textured till-Bt and the finer-textured loess-Bt. This similarity suggested a dominant effect of pedogenesis and a minor effect of the parent material on the roughness levels of structural surfaces in the Bt-horizons. An expected "smoothing" effect of burrow wall surfaces by earthworm activity was not reflected in the roughness values compared to those of uncoated cracks at the chosen spatial scale. However, for root channel walls from one loess-Bt, the roughness was reduced as compared to that of other structures. These results suggest that the surface roughness of the structural surface types should separately be considered when describing preferential flow and macropore-matrix exchange or analysing root growth, microbial habitats, and colloidal transport in structured soils. The confocal laser scanning microscopy technique was found useful for characterizing the roughness of intact structural surfaces.

Key words: clay coatings / earthworm burrow / micro-topography / organo-mineral coating / preferential flow / soil structure

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1 Introduction

During preferential flow events in structured soils, the transport and exchange of water and reactive solutes between macropores and the soil matrix is mostly restricted to the surfaces of preferential flow path structures. These structures are part of the macropore system consisting of cracks, inter-aggregate pore spaces, and biopores. The micro-topography of these macropores in terms of pore shapes, geometry, and roughness is crucial for describing preferential flow. The roughness of structural surfaces also can be an important characteristic for microbial habitats in structured soil (e.g., *Vinther* et al., 1999). The movement of water and solutes through structured soils and the exchange between these flow domains can be described by a dual-permeability model (*e.g.*, *Gerke* and *van Genuchten*, 1993) coupling both preferential and matrix transport. The mass transfer between these two domains is quantified by exchange terms which require physico-chemical and structural properties of the preferential flow path surfaces (*Gerke*, 2012). Amongst the latter ones, the surface-to-volume ratio of soil aggregates, *i.e*., the specific surface of preferential flow paths in relation to the bulk soil volume is of importance and is affected by the roughness of the structural surfaces.

Luvisols are characterized by a distinctive crack system, especially in the subsoil, which is connected to the biopores. As a specific attribute, cracks in Luvisol subsoil are often covered by clay or clay-organic coatings resulting from colloidal transport from the topsoil to the subsoil (Bt-horizon) where the colloids were immobilized. These coatings have different physcio-chemical properties, compared to the uncoated soil matrix or uncoated cracks, *i.e.*, a lower or higher cation exchange capacity (*Ellerbrock* and *Gerke*, 2004), a lower wettability (*Leue* et al., 2013, 2015), and a reduced hydraulic conductivity or diffusivity (*Gerke* and *Köhne*, 2002; *Köhne* et al., 2002).

* Correspondence: Dr. M. Leue; e-mail: Martin.Leue@zalf.de

However, the relatively fine texture of the illuviated material implies that crack coatings have smoother surfaces compared to uncoated cracks, which may promote effects on the water movement through these macropores and on the mass exchange between macropores and the porous soil matrix.

The surfaces of biopores can be coated by organic material as well, deriving either from root residues and exudates or from earthworm mucilage and casts and from incorporated litter. Again, the composition and spatial distribution of the organic matter (OM) affects the physico-chemical surface properties of the biopores. In contrast to the cracks, the surfaces of biopores may not only be smoothed by fine-textured OM but also by the radial pressure of roots (*Ruiz* et al., 2015) and earthworms (*Rogasik* et al., 2014). In earthworm burrows, the longitudinal movement of the earthworms results in a "smearing" of soil and cast material (*Jégou* et al., 2001; *Ellerbrock* et al., 2009). These processes lead to smoother surfaces of earthworm burrows in relation to the surrounding soil, as shown, but not parameterized, by *Jégou* et al. (2001), using scanning electron microscopy (SEM). However the well-known high microbial activity along earthworm burrows (*Vinther* et al., 1999; *Bundt* et al., 2001) may increase the roughness of the burrow walls due to a perforation by small lateral pores.

Among the technologies to measure surface roughness or micro-topography of intact surfaces in soils, noncontact laser techniques are used at a broad spatial scale from landscapes to soil micro-aggregates. The confocal laser scanning microscopy (CLSM) is able to analyze the micro-topography of regions at the mm to sub-mm scale. The use of a laser beam allows for measurements of elevations at the sample surface, which is not possible by other microscopic techniques such as SEM. The combination of laser and optical scans from different elevation levels enables the display of combined highly-resolved micro-photographs and elevation measurements. Well-established in biotechnology (*Paddock*, 1999), CLSM provides ideal basic conditions to describe and quantify the surface roughness and micro-topography by standardized parameters.

We hypothesize that the surface roughness of crack and biopore surfaces in Bt-horizons of Luvisols developed from loess and glacial till as parent material differs with respect to texture, coatings, and type of the structural surface as follows: (1) The texture of the structural surface depends on that of the parent material, *i.e*., structural surfaces of finer textured soils (loess) are smoother than surfaces found in coarser textured soils (till). (2) Crack coatings and biopore surfaces have smoother surfaces than uncoated cracks. (3) This difference between the surface types is larger for coarser textured (till) than for finer textured (loess) soils.

The objective of this study was to characterize the surface roughness of intact structural surfaces, *i.e*., cracks with and without clay-organic coatings, earthworm burrows, and root channels, from Luvisol Bt-horizons developed on loess and glacial till by confocal laser scanning microscopy. The obtained roughness values were evaluated with respect to the texture of the parent material and the surface types in order to describe micro-topographic differences between the structural surfaces as a result of the formation of structured soils.

2 Material and methods

2.1 Soils and sampling

Soil samples were collected from four Haplic Luvisols (*WRB*, 2006) located in SW (Simmringen) and NE Germany (Holzendorf) and in the N Czech Republic (Hnevceves) and from an Albic Luvisol located in NE Germany (Rowa). The Luvisols developed from loess (*L*) and from glacial till (*T*), under arable (*a*) and forest (*f*) land use (Table 1). All Bt-horizons were well-structured by cracks, fissures, and earthworm burrows, which could potentially serve as preferential flow paths. At each site, soil pits of edge lengths of 120 cm × 120 cm were excavated, from where larger soil blocks of approximately 15 cm × 20 cm

Table 1: Basic site and soil information. P–mean annual precipitation, T–annual mean temperature, SOC–soil organic carbon. For further details, see *Kaiser* et al. (2010) for Simmringen and Rowa, *Kodešová* et al. (2011) and *Fér* and *Kodešová* (2012) for Hnevceves, and *Leue* et al. (2013).

Site name	Site abbrev.	Geogr. position / °E, °N	Altitude a.s.l. / m	P / mm year^{-1}	T / °C	Parent material, land use	Depth / cm	pH	Clay / %	Silt / %	Sand / %	SOC / %
Simmringen	L1(a)	09°52′57′′, 49°34′47′′	338	577	9.4	loess, arable	60–80	7.2	35.4	63.2	1.4	0.52
Simmringen	L1(f)	09°52′57′′, 49°34′47′′	338	577	9.4	loess, forest	60–70	4.6	34.3	63.0	2.7	0.43
Hnevceves	L2(a)	15°43′03′′, 50°18′47′′	273	618	8.5	loess, arable	60–80	6.6	34.6	56.2	9.2	0.44
Holzendorf	T1(a)	13°47′11′′, 53°22′45′′	48	501	8.7	glacial till, arable	40–60	6.4	16.5	35.0	48.5	0.31
Rowa	T2(f)	13°16′20′′, 53°29′47′′	90	532	7.9	glacial till, forest	80–100	4.7	16.8	24.4	58.8	0.24

× 20 cm edge lengths were cut out of the Bt-horizon using a spade. In the laboratory, smaller subsamples (aggregates and clods) of about 5 to 10 cm edge lengths were manually separated from the larger soil blocks of each site to obtain intact crack and biopore surfaces. The sides of the subsamples were fixed with tinfoil to prevent clods from cracking. Fine plant roots growing along cracks and earthworm burrows were removed. The stabilized final samples were fixed on an aluminum plate and dried over silica gel in a desiccator (*Leue* et al., 2010).

At the intact surfaces that were prepared from the larger clods, the following surface types were identified by visual inspection: uncoated cracks (CS), coated cracks (CS+C), earthworm burrow walls (EB), and root channels (RC). Root channels of 5–8 mm in diameter could be identified only at the forest site L1(f). Along all samples, although fine plant roots (< 0.5 mm in diameter) were observed, no corresponding root channels could be identified at the surface of cracks. Between 6 and 11 regions of approximately one mm^2 in area of each surface type per site were selected for the surface-roughness measurements.

2.2 Confocal laser scanning microscopy (CLSM)

The roughness of the intact structural surfaces was measured using a three-dimensional confocal laser scanning microscope (CLSM; Keyence VK-X100K; Keyence Company, Osaka, Japan). Regions of interest (ROI) with edge lengths of 0.675 mm × 0.506 mm (*i.e.*, 0.342 mm^2) in length and width (x- and y- dimensions) were scanned with a 400-fold magnitude. The resolutions of the measurements were 0.05 μm in x- and y-dimensions and 0.02 μm in z-dimension (vertical direction). For each selected ROI, the upper and lower limit of the laser scan was focused to the highest and the lowest elevation levels within the ROI.

The CLSM provided information on the color (as visible to the naked eye; Fig. 1), the reflected laser intensity, and the sample surface micro-topography (*i.e.*, the relative elevation at each pixel; Fig. 2) of the scanned region (x- and y- dimensions). The Keyence software provided several algorithms for smoothing the z-values. However, the data shown here were not smoothed in order to assess the original surface's roughness and to keep the possibility to easily compare the data with other laser scan measurements. The overall slope of the ROI, which resulted from a non-horizontal positioning of the sample under the laser ocular, was corrected using an automated curved secant procedure that was an implemented in the software.

The software also provided the possibility to exclude very high and very low elevation values (z-values) from the roughness calculation. These values can be measured either at perpendicular pores (holes) and shrinking fissures (causing extremely low or negative z-values) or at directly reflecting mineral particles or very smooth surface regions (causing extremely high z-values). However, the data shown here were not threatened by this procedure since (1) perpendicular pores are inherent elements of the structural surfaces and thereby affect the surface's roughness, (2) the occurrence of shrinkage fissures was avoided by the small ROI size, and (3) direct reflections were observed only in a few ROI but with small spatial extensions (*e.g.*, Fig. 2, left corner of the figure T1(a)–coated cracks).

Statistical analyses were carried out using SPSS (Version 22, IBM Corp., USA). Differences between the mean values of the parameters were tested at the significance levels of $P < 0.05$ using the Kruskal–Wallis analysis of variance fol-

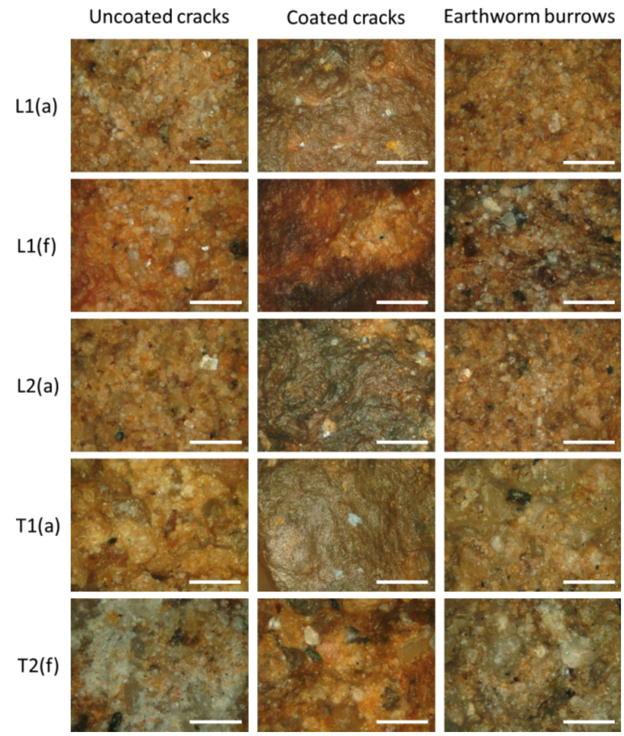

Figure 1: Micro-photographs of ROI at uncoated (left column) and coated (middle column) cracks and at earthworm burrows (right column) from the Bt-horizon of five Luvisols (L–loess, T–till, a–arable, f–forest). Scale bar: 200 μm.

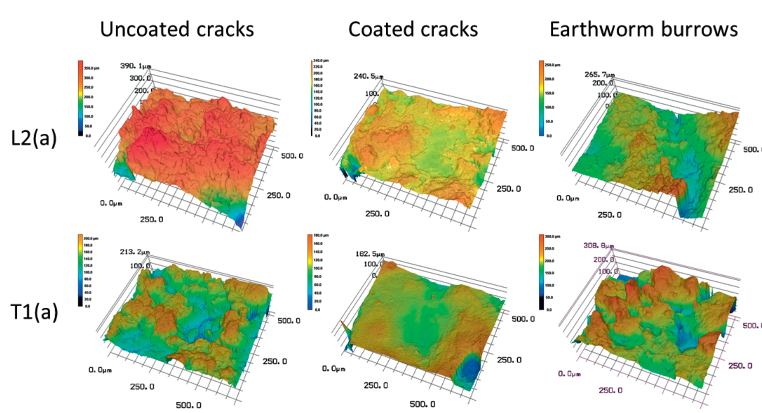

Figure 2: Spatial distribution of height values of ROI at intact structural surfaces of a loess-derived [L2(a)] and a till-derived Bt-horizon [T1(a)].

lowed by pairwise multiple comparisons (Dunn's post hoc test).

2.3 Roughness parameters

The following roughness parameters were calculated from the surface elevations within the ROI (Fig. 2) as implemented in the analysis software (VK-H1XAD, Keyence Company, Japan) of the CLSM (*ISO 4287:1997*, 1997; *JIS B 0601-2001*, 2001):

The root-mean-squared roughness (R_q; dimension: µm) is the square-root of the mean squared elevation values of all measurement points (N) within the ROI:

$$R_q = \sqrt{\frac{1}{N}\sum_{n=1}^{N} Z_n^2}. \tag{1}$$

The curvature (or kurtosis; dimensionless; R_{cu}) as a measure of 'peakedness' is the mean value of the fourth power of the elevations divided by the fourth power of Z of every measurement point within the ROI:

$$R_{cu} = \frac{1}{R_q^4}\left[\frac{1}{N}\sum_{n=1}^{N} Z_n^4\right]. \tag{2}$$

As a non-ISO standard parameter, the ratio between surface area and base area (R_A; dimensionless), of the ROI was calculated with the Keyence software tool "Volume / Area" using the "Surface area / Area" option as:

$$R_A = \frac{Surface\ area\ (ROI)}{Base\ area\ (ROI)}. \tag{3}$$

The surface area of the ROI was defined as the overall area while the base area was the area of the ROI (0.342 mm^2). Note that the volumes of this calculation could not be used since the sample volume was modified by the preparation procedure and hence was arbitrary.

3 Results

3.1 Roughness of structural surfaces of Luvisols from different parent materials

For three among five Luvisol Bt-horizons, coated cracks (CS+C) revealed smaller root-mean-squared roughness (R_q) values as compared to uncoated cracks (CS) and earthworm burrows (EB; Table 2). For all Luvisols, coated cracks showed smaller ratios between surface and base area of the ROI (R_A) as compared to uncoated cracks and earthworm burrows. However, the differences were found to be statistically signifi-

Table 2: Root-mean-squared roughness (R_q), curvature (R_{cu}), and ratio between surface area and base area (R_A) of intact structural surfaces from the Bt-horizons of five Luvisols. Mean values and standard deviations (in brackets). Significant differences between the mean values within the same site were denoted with *c* (crack coatings *vs*. earthworm burrows), *d* (uncoated cracks *vs*. root channels), and *f* (earthworm burrows *vs*. root channels).

Site	L1(a)	L1(f)	L2(a)	T1(a)	T2(f)
Parent material	loess	loess	loess	glacial till	glacial till
Land use	arable	forest	arable	arable	forest
R_q / µm					
uncoated cracks	23.8 (8.8)	37.7 (18.5)^d	24.2 (6.4)	35.9 (7.2)	34.2 (10.0)
coated cracks	13.4 (2.2)	23.9 (8.2)	27.3 (10.7)	22.0 (6.1)	33.3 (6.8)
earthworm burrows	26.4 (8.7)	33.2 (10.2)^f	25.9 (5.1)	35.4 (6.5)	46.3 (18.6)
root channels		12.0 (5.7)$^{d, f}$			
R_{cu} / –					
uncoated cracks	3.73 (0.87)	5.38 (2.89)	3.45 (0.70)	3.33 (0.66)	3.36 (0.53)
coated cracks	4.02 (1.51)	4.53 (2.02)	5.10 (2.21)	5.13 (2.30)	3.72 (1.11)
earthworm burrows	5.06 (2.18)	3.69 (1.07)	3.68 (0.91)	3.49 (0.45)	3.54 (1.09)
root channels		10.05 (6.15)			
R_A / –					
uncoated cracks	3.78 (0.43)	3.92 (0.81)	3.90 (0.26)	4.03 (0.48)	4.48 (0.62)
coated cracks	3.16 (0.28)	3.97 (0.57)	3.58 (0.60)	3.37 (0.45)	3.97 (0.35)^c
earthworm burrows	3.66 (0.27)	4.58 (0.61)	3.84 (0.55)	3.76 (0.47)	5.60 (0.91)^c
root channels		3.55 (0.97)			

cant only between the R_q values of uncoated cracks and earthworm burrow walls in relation to root channels (RC) of the forest site L1(f) and between the R_A values of coated cracks and earthworm burrows for the site T2(f). The curvature (R_{cu}) of root channels from the site L1(f) was found to be higher than for uncoated cracks and earthworm burrows from the other sites; however, this difference was not statistically significant when comparing samples only from the site L1(f). The R_q values were positively correlated with the R_A values (R = 0.67; p = 0.01), and weakly negatively correlated with the R_{cu} values (R = –0.19; p = 0.05); between R_{cu} and R_A no correlation was found.

For both Luvisol parent materials, loess and till, the mean values of R_q and R_A were lower for coated cracks than for uncoated cracks and earthworm burrows (Fig. 3a, c). Among the loess samples, the R_q and R_A values were lower and the R_{cu} values were higher for root channels compared to uncoated cracks, coated cracks, and earthworm burrows (Fig. 3b). The loess-derived surfaces revealed slightly lower R_q and R_A levels and a slightly higher R_{cu} level compared to the till-derived ones.

3.2 Roughness parameters of structural surface types

The roughness parameters of the different structural surfaces types in general, aggregated from both the loess- and till-derived Luvisols, are shown in Fig. 4. Corresponding to the results of the single sites (Table 2) and the parent materials (Fig. 3), the coated cracks showed lower R_q and R_A values than uncoated cracks and earthworm burrows. Uncoated cracks and earthworm burrows revealed nearly the same levels for all parameters. The root channels from the loess-derived Bt of L1(f) showed significantly lower R_q and slightly lower R_A values than the other surface types. In contrast, the R_{cu} values of root channels were significantly higher compared to uncoated and coated cracks and earthworm burrows which showed almost the same R_{cu} level but with decreasing standard deviations (Fig. 4b).

The R_q values of earthworm burrows and the R_A values (Table 2) of uncoated cracks were correlated with the organic carbon (OC) content (Table 1) of the Bt-horizons (R = 1.0, Spearman, p = 0.01). The R_{cu} values of the earthworm burrows (Table 2) were correlated positively (R = 0.9) with the silt and clay contents (Table 1) and negatively with the sand content (R = –0.9) of the Bt-horizons (Spearman, p = 0.05).

4 Discussion

4.1 Uncoated *vs.* coated cracks

The intact surfaces of coated cracks were smoother than the surfaces of uncoated cracks as shown for most of the single sites (Table 2), for the aggregated data levels of the parent material (Fig. 3), and the structural surface types (Fig. 4). These results can also be seen at the micro-photographs of the ROI (Fig. 1). The roughness data confirm the well-known fact that crack coatings consist of finer-textured clay and or-

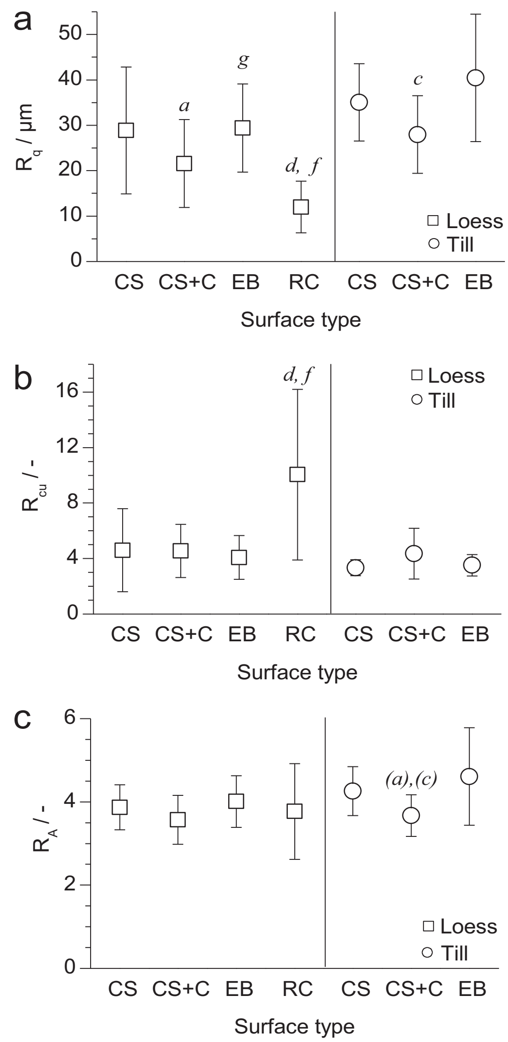

Figure 3: Roughness parameters of uncoated cracks (CS), coated cracks (CS+C), earthworm burrows (EB), and root channels (RC), aggregated at the level of the Luvisol parent material (squares–loess, circle–till). Significant differences between the mean values denoted with *a* (uncoated *vs*. coated cracks), *c* (coated cracks *vs*. earthworm burrows), *d* (uncoated cracks *vs*. root channels), *f* (earthworm burrows *vs*. root channels), and *g* (loess-derived EB *vs*. till-derived EB). Letters in brackets denote a significance level of P = 0.1 (instead of P = 0.05).

ganic material (*e.g*., *Leue* et al., 2016), which can reduce the exchange of water and solutes between the macropores and the porous soil matrix (*Gerke* and *Köhne*, 2002; *Köhne* et al., 2002).

The slightly higher roughness (R_q and R_A values) of the structural surfaces from the till-derived Bt-horizon, with significant differences only between the R_q values of the earthworm burrows (Fig. 3), can be attributed to the coarser texture of the till compared to the loess (Table 1). Such effect of the parent material of soil formation on the roughness of those structural surfaces, which are not covered by clay-organic coatings, is also underlined by the correlation between the surface roughness of uncoated cracks and earthworm burrows (Table 2) and the texture of the bulk soil of the Bt-horizons (Table 1).

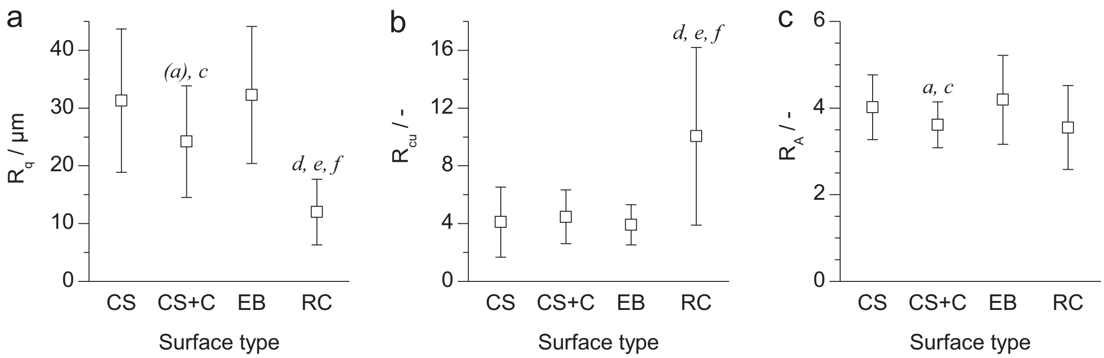

Figure 4: Roughness parameters of uncoated cracks (CS), coated cracks (CS+C), earthworm burrows (EB), and root channels (RC), aggregated at the level of structural surface types from both the loess- and till-derived Luvisols. Significant differences between the mean values denoted with *a* (uncoated *vs*. coated cracks), *c* (coated cracks *vs*. earthworm burrows), *d* (uncoated cracks *vs*. root channels), *e* (coated crack *vs*. root channels), and *f* (earthworm burrows *vs*. root channels). Letters in brackets denote a significance level of $P = 0.1$ (instead of $P = 0.05$).

Comparing the texture of bulk soil and clay-organic coatings, the coarser textured till-Bt should principally reveal higher differences between the roughness parameters of uncoated and coated cracks than the finely textured loess-Bt. However, such relationship could only be found for the R_A values of the till samples which showed slightly higher differences between coatings and uncoated crack surfaces compared to the loess samples (Fig. 3).

The correlation between the surface roughness (Table 2) and the mean OC content of the Bt-horizons (Table 1) illustrates the effect of a continuous accumulation of organic matter in the Bt-horizons that takes place predominantly at crack surfaces (*e.g*., *Kodešová* et al., 2011). More and thicker coatings imply higher mean OC contents of the Bt-horizon and lower the surface roughness during the Luvisol genesis. However, due to the relatively small number of investigated sites ($n = 5$), the correlations between the roughness parameters (Table 2) and the Bt-bulk soil parameters (Table 1) have to be considered as preliminary results.

4.2 Earthworm burrow walls

In contrast to findings of *Jégou* et al. (2001), who showed smooth earthworm burrows by SEM visualization, the roughness parameters R_q, R_{cu}, and R_A of earthworm burrows did not show significantly smaller values and standard deviations compared to uncoated cracks [Table 2, Figs. 1 (left and right columns), 3 and 4]. While in *Jégou* et al. (2001) the micrographs suggested relatively smooth earthworm burrow surfaces, the roughness was not quantified and the spatial resolution of the SEM micrographs was coarser as compared to that of the CLSM method used here. However, also at a much coarser macroscopic scale (*i.e*., as visible to the naked eye), earthworm burrows appear smoother than crack surfaces. We explain this difference by the scale dependence of the of roughness characteristics, *i.e*., the spatial resolution of the measurement. At the sub-mm scale (0.342 mm^2 ROI, 400-fold magnitude), the roughness of the earthworm burrow walls equaled that of uncoated cracks since the earthworms do not modify the soil texture in the same way as the illuviation of finer-textured clay-organic particles do at crack and aggregate surfaces. At this relatively small spatial scale, the different processes of earthworm burrow formation and destruction, such as radial pressure (*Rogasik* et al., 2014), smearing of casts and soil material (*Jégou* et al., 2001), and perforation of the walls by small lateral pores resulting from increased microbial activity (*Vinther* et al., 1999; *Bundt* et al., 2001), do not significantly modify the general texture-dominated surface roughness.

Earthworm burrow walls are known to be coated with OM (*e.g*., *Edwards* et al., 1992; *Stehouwer* et al., 1993), which could be monitored by infrared spectroscopy (*Ellerbrock* et al., 2009; *Leue* et al., 2013). The similarity of the roughness of burrows and uncoated cracks implies that the coatings in fact cover the particle surfaces but do not alter the textural composition of the burrow walls. The OM coatings of particles in the burrow walls (but not the drilosphere itself) are thinner than the clay-organic coatings along cracks. In contrast to burrow walls, clay-organic coatings along cracks (Fig. 1, middle column) developed by filling the pore space between larger particles with fine-textured illuviated material, thereby creating a smoother surface with a modified texture (Fig. 4), which can reduce water absorption by soil aggregates (*Kodešová* et al., 2011; *Fér* and *Kodešová*, 2012).

4.3 Root channels

Root channels could be identified only at samples from the forested Luvisol L1(f) and were principally smooth (lowest R_q values), but revealed a distinctive curvature (highest R_{cu} values among all measurements; Fig. 4). Similar to the earthworm burrows, the rhizosphere soil is compressed by the radial pressure of the growing roots (*Ruiz* et al., 2015). However, roots can exert pressures up to 100-times larger than maximal earthworm pressures (*Bengough* and *Mullins*, 1990; *Ruiz* et al., 2015). Corresponding to the earthworm burrowing, the root growth does not alter the texture of the soil. In contrast to the earthworms, the much higher pressure of the roots may have rearranged the particles along the root channel wall which, in addition with the smoothing effect of OM-rich coat-

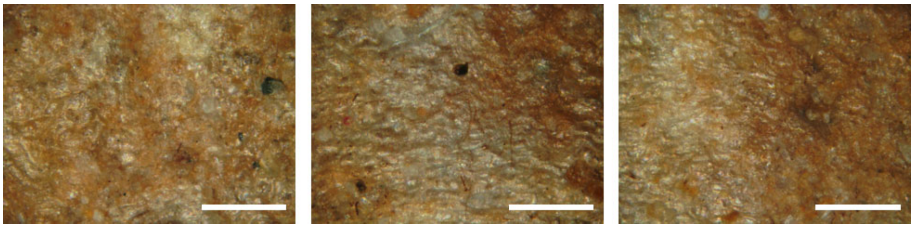

Figure 5: Micro-photographs of ROI at root channel walls from the Bt-horizon of a loess-derived Luvisol under forest land use [L1(f)]. Scale bar: 200 μm.

ings along the root channel walls (Fig. 5; *Leue* et al., 2013), leads to lower R_q values. However the small thickness of the coatings preserves effects of the underlying soil particles and results in high curvature values of the root channels (Table 2, Figs. 3 and 4).

4.4 Methodological issues

The high R_{cu} values of the root channels and the weak or missing correlations between R_{cu}, R_q and R_A indicate that the curvature of the structural surfaces is determined neither by R_q nor by the R_A. Surfaces with a small mean roughness can, but do not have to, feature high curvature and *vice versa*.

One methodological problem of clayey soil samples was the occurrence of perpendicular shrinkage fissures at the surface, which could simulate a higher surface roughness than actually existing ('artifacts'). Previous roughness data from own measurements at a larger ROI of 1.350 mm × 1.012 mm (1.366 mm^2) were found to be much more affected by such artifacts, which caused extremely high standard deviations of the roughness parameters (data not shown). We avoided this problem by choosing a relatively small ROI of 0.675 mm × 506 mm (0.342 mm^2), which allowed to measure surface regions without fissures that are oriented perpendicularly to the surface of the ROI. Measurements of field-moist samples could be an alternative to the treatment of dry samples used here in order to reduce the risk of the formation of additional shrinkage cracks after the soil sampling. Despite the small ROI size, we aimed at considering roughness parameters that are related to the macroscopic scale of preferential flow descriptions (*Gerke*, 2012) by averaging the roughness values of up to 11 ROI per surface type of each Luvisol Bt-horizon. However, the small ROI size does not comprise the complexity of the macropores and is only one piece to characterize the soil structure.

Another issue was the handling of perpendicular pores at the intact surfaces. On the one hand, these 'holes' can cause shading effects of the optical and laser radiation, which may lead to implausible low elevation values. On the other hand, these pores are inherent elements of the structural surfaces and thereby also affect the roughness of the surfaces under in-situ conditions. We included the regions of the perpendicular pores in the calculation of the roughness parameters since we could observe only a few extreme or frequent shading effects. The exclusion of these pore regions inside the ROI from the roughness calculation would lead to considerably lower R_q, R_{cu}, and R_A values with lower standard deviations (data not shown). The similar effect could be obtained by smoothing the elevation data (z-values) with the algorithms implemented in the software. However, the inter-comparison of roughness values from different soils would be complicated by the application of such procedures. Note that a pre-processing of the data (*i.e.*, smoothing and excluding of extreme z-values) would increase the statistical significance of the differences between mean values by reducing the spread of the roughness parameters. However, the relative differences in the roughness parameters between the samples or the sample groups would remain relatively similar.

5 Conclusions

The roughness of intact structural surfaces in Luvisol Bt-horizons developed from loess and glacial till was characterized by relatively small (0.342 mm^2) scale measurements using 3D confocal laser scanning microscopy (CLSM). The mean squared roughness (R_q) and the curvature (R_{cu}) as ISO parameters, and the ratio between the surface and the base area (R_A) of the measured regions of interest on the sample surfaces were used to parameterize the surface roughness of cracks with and without clay-organic coatings, as well as of earthworm burrow walls, and root channels. The following conclusions were obtained:

(1) The roughness data measured for the selected sites and at a sub-mm spatial scale suggested that different parent materials (here: loess and glacial till) did only weakly influence the roughness of the intact structural surface types. Obviously the effect of the parent material on the roughness of the structures is overlaid by different soil-forming processes, *e.g.*, clay illuviation and root growth.

(2) The illuviation of fine-textured clay-organic coatings reduced the surface roughness of cracks in Luvisol Bt-horizons. In particular the reduced ratio between surface region and base region implies reduced surface-to-volume ratios of soil aggregates as a consequence of the colloid migration. This reduction of the specific surface area of preferential flow paths in relation to the bulk soil may lead to accelerated water movement through the crack system during preferential flow events. An effect of this roughness-induced acceleration of macropore flow in coated cracks on the mass exchange with the soil matrix may be expected due to shorter contact times.

(3) The similarities between uncoated cracks and earthworm burrows implied that smoothing by earthworms is not relevant at the investigated spatial scale. In contrast, root channels of

persistent roots (under forest) revealed a lower surface roughness as compared to earthworm burrows as result of higher pressures that altered the arrangement of soil particles along the root channel walls. The relatively high roughness of earthworm burrows seems to provide an increased number of micro-habitats.

The results may help to develop a more detailed knowledge on soil structural properties. In particular, the results suggest that different kinds of macropores, such as cracks with and without coatings, earthworm burrows, and root channels, should be distinguished with respect to their surface roughness when modelling preferential flow in Luvisols, analyzing root growth, microbial habitats, and colloidal transport in structured soils. The confocal laser scanning microscopy technique was found useful for characterizing the roughness of intact structural surfaces at the sub-mm scale with a μm-scale elevation resolution and could similarly be applied, for instance, to quantify conditions for micro-habitats and effects of root growth along macropores.

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